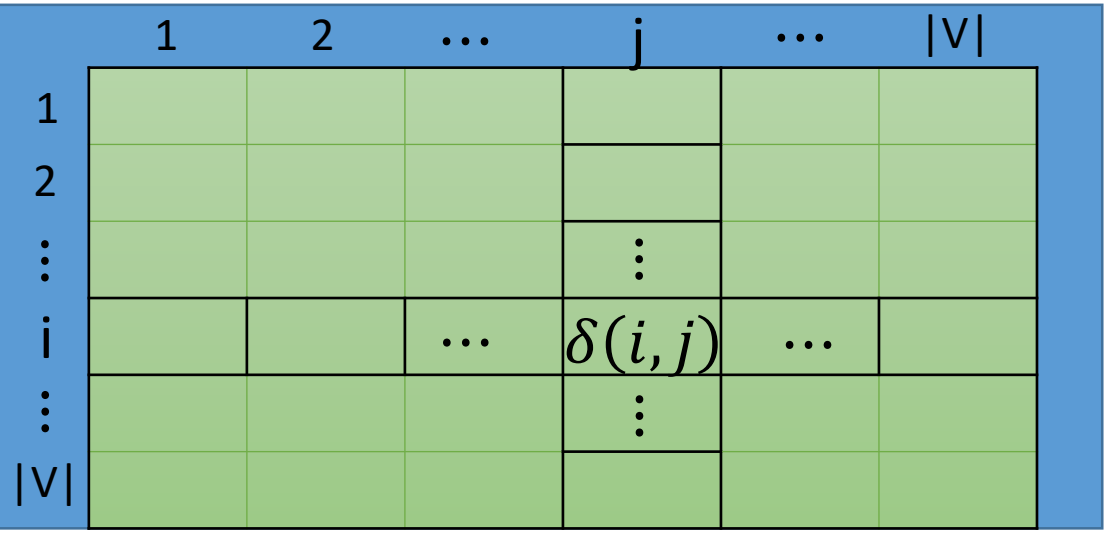
APSP- All Pairs Shortest Paths

We can solve this problem using **Belman-Ford** by executing it from *each vertex*

The cost of this solution is:
$\Theta(|V|^2|E|) = O(|V|^4)$

The **output** required for this problem is a 2-dimensional matrix of size $\Theta(|V|^2)$



Dynamic Programming

Development of Dynamic Programming Algorithm

1. Optimal solution structure specification
2. Recursive formalization of optimal value
3. Bottom-up computation of optimal value (using a table)
4. Optimal solution construction using the computed data

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Floyd-Warshal's Algorithm Cost

Command Number	Cost
1	$\Theta(\|V\|)$
2	$\Theta(\|V\|^2)$
3-6	$\Theta(\|V\|^3)$
7	$\Theta(1)$
Total	$\Theta(\|V\|^3)$

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Floyd-Warshal's Algorithm Correctness

Claim:

Let $G = (V, E)$ be a weighted digraph with $w: E \to \mathbb{R}$.

After the **k-th** iteration of the for-loop in line 3 of Floyd-Warshal's algorithm, it holds that:

for every $i, j \in V$, $d_{i,j}^{k}$ is the weight of a shortest path from i to j that may use intermediate vertices from the set: $\{1, \cdots, k\}$.

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Floyd-Warshal's Algorithm Correctness

Proof (of claim): By induction on k.

Base Case: **k=0**

In this case, k is less than the lowest valid vertex value.
Therefore, no intermediate vertices can be used before the first iteration.

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Proof (of claim): By induction on k.

Base Case: **k=0**

In this case, k is less than the lowest valid vertex value.
Therefore, no intermediate vertices can be used before the first iteration.
Indeed, before the first iteration $d_{i,j}^{0}$ values are defined in line 2 to be the weight of direct edge from i to j (if exists), for every $i, j \in V$.
Therefore, $d_{i,j}^{0}$ is indeed the weight of shortest direct path.

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Proof (cont.):

Induction step:

Assume by induction that after the (k-1)-th iteration of line 3, for every i, j $\in V$, $d_{i,j}^{k-1}$ is the weight of a shortest path from i to j that may use intermediate vertices from the set: $\{1, \cdots, k-1\}$.

We prove that after the k-th iteration of line 3, for every i, j $\in V$, $d_{i,j}^{k}$ is the weight of a shortest path from i to j that may use intermediate vertices from the set: $\{1, \cdots, k\}$.

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Proof (cont.):

There are two possibilities for a shortest path from i to j that may use intermediate vertices from the set $\{1, \cdots, k\}$:

1. either it does not use the vertex k, or
2. it uses the vertex k.

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Proof (cont.):

There are two possibilities for a shortest path from i to j that may use intermediate vertices from the set $\{1, \cdots, k\}$:

1. either it does not use the vertex k, or
2. it uses the vertex k.

In case 1, the shortest path from i to j that may use intermediate vertices from the set $\{1, \cdots, k\}$ is actually a shortest path from i to j that uses intermediate vertices only from the set $\{1, \cdots, k-1\}$. Therefore, by induction hypothesis its value is $d_{i,j}^{k-1}$.

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Floyd-Warshal's Algorithm Correctness

Proof (cont.):

In case 2, shortest such path has 2 parts:

By sub-paths optimality of shortest paths

Which intermediate vertices used in sub-paths ?

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Floyd-Warshal's Algorithm Correctness

By sub-paths optimality of shortest paths

Which intermediate vertices used in sub-paths ?

Proof (cont.):

In case 2, shortest such path has 2 parts:

By induction hypothesis, we have that $d_{i,k}^{k-1}$ contains the weight of shortest path from i to k that may use intermediate vertices from the set $\{1, \cdots, k-1\}$ after the (k-1)-th iteration. Similarly, $d_{k,j}^{k-1}$ contains the weight of shortest path from k to j that may use intermediate vertices from the set $\{1, \cdots, k-1\}$ after the (k-1)-th iteration.

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Floyd-Warshal's Algorithm Correctness

Proof (cont.):

By the definition of path-weight the total path weight is the sum.

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Floyd-Warshal's Algorithm Correctness

Proof (cont.):

By the definition of path-weight the total path weight is the sum.

Since the computation in line 6 assigns to $d_{i,j}^{k}$ the better weight of the two cases 1 and 2, we have that after the k-th iteration of line 3, for every i, j $\in V$, $d_{i,j}^{k}$ is the weight of a shortest path from i to j that may use intermediate vertices from the set: $\{\mathbf{1}, \cdots, \boldsymbol{k}\}$.

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Floyd-Warshal's Algorithm Correctness

Conclusion:

Let $G = (V, E)$ be a weighted digraph with $w: E \to \mathbb{R}$.
When Floyd-Warshal algorithm halts, we have that:
for every i, j $\in V$, $D^n[i, j] = \delta(i, j)$.

Proof (of conclusion):

Take k=n in the last claim, and conclusion follows (why?).

What is the space complexity ?

How many tables algorithm needs?